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# **Evaluating Naïve Bayes, kNN, SVM, and Decision Trees for Shuttle Data Classification**

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# 1 Naïve Bayes Classifier

## 1.1 Performance Overview

The Naïve Bayes classifier was systematically evaluated across three distinct parameter configurations, revealing significant variations in classification performance. Table 1 presents the comprehensive metrics:

Table 1: Naïve Bayes classification performance with varying parameters

| Parameters                   | Accuracy (%) | Error (%) | Cohen’s $\kappa$ | Correct | Incorrect |
|------------------------------|--------------|-----------|------------------|---------|-----------|
| $p = 0.0001, \sigma = 0.001$ | 76.14        | 23.86     | 0.468            | 11,040  | 3,460     |
| $p = 0.001, \sigma = 0.0001$ | 55.80        | 44.20     | 0.252            | 8,091   | 6,409     |
| $p = 0.005, \sigma = 0.01$   | 75.83        | 24.17     | 0.476            | 10,996  | 3,504     |

The optimal configuration ( $p = 0.0001, \sigma = 0.001$ ) achieved 76.14% classification accuracy with a Cohen’s  $\kappa$  of 0.468, indicating *moderate agreement* beyond random chance according to Landis and Koch’s benchmark scale. This represents a 36.7% relative improvement over the worst-performing configuration ( $p = 0.001, \sigma = 0.0001$ ), which yielded only 55.8% accuracy.

Three key observations emerge:

- **Parameter Sensitivity:** The 20.34 percentage point accuracy drop between the best and intermediate configurations demonstrates the algorithm’s sensitivity to probability threshold ( $p$ ) adjustments, particularly in the  $10^{-4}$  to  $10^{-3}$  range.
- **Error Analysis:** The optimal configuration’s 23.86% error rate decomposes into:
  - *Class-dependent errors:* Primarily from minority class misclassifications (as shown in Section 3.2)
  - *Model bias:* Inherent limitations from the independence assumption
- **Statistical Significance:** The  $\kappa$  values (0.252–0.476) suggest that while the model performs substantially better than random guessing ( $\kappa = 0$ ), there remains considerable room for improvement toward excellent agreement ( $\kappa > 0.6$ ).

## 1.2 Confusion Matrix Analysis

Table 2 shows the confusion matrix for the best-performing configuration:

Table 2: Confusion matrix for Naïve Bayes ( $p = 0.0001, \sigma = 0.001$ )

| Actual / Predicted | 1     | 2     | 3     | 4     | 5      | 6 | 7    | Recall (%) |
|--------------------|-------|-------|-------|-------|--------|---|------|------------|
| 1                  | 9,347 | 1,097 | 103   | 138   | 0      | 0 | 711  | 82.02      |
| 2                  | 0     | 12    | 0     | 0     | 0      | 0 | 0    | 100.00     |
| 3                  | 1     | 8     | 25    | 0     | 0      | 0 | 9    | 58.14      |
| 4                  | 556   | 697   | 0     | 973   | 0      | 0 | 0    | 43.71      |
| 5                  | 1     | 134   | 0     | 0     | 682    | 0 | 0    | 83.48      |
| 6                  | 1     | 1     | 0     | 1     | 0      | 0 | 0    | 0.00       |
| 7                  | 1     | 1     | 0     | 0     | 0      | 0 | 1    | 33.33      |
| Precision (%)      | 94.35 | 0.62  | 19.53 | 87.50 | 100.00 | – | 0.14 |            |

The confusion matrix highlights several important aspects of the model’s performance under the parameter configuration  $p = 0.0001$ ,  $\sigma = 0.001$ . Class 1, being by far the most frequent in the dataset, achieved both high recall (82.02%) and very high precision (94.35%). This indicates that the model is heavily biased toward predicting this class, likely contributing to the misclassification of several minority classes.

Class 2 achieved perfect results with both precision and recall at 100%; however, it only appeared 12 times in the test set, limiting its influence on overall performance. Class 3, in contrast, showed much weaker results, with a recall of 58.14% and precision of only 19.53%, revealing that the model had trouble distinguishing this class from others such as 2 and 7.

Class 4 presented a recall of 43.71% and a relatively high precision of 87.50%, suggesting that although the model was fairly reliable when it predicted Class 4, it frequently failed to identify it correctly. Class 5 performed strongly, with a perfect precision of 100% and high recall (83.48%), indicating that the model learned to distinguish this class well.

Class 6, despite being present in the test set, was never correctly predicted (0% recall), pointing to a clear failure in learning from this underrepresented class. Similarly, Class 7 achieved only 33.33% recall and had a very low precision of 0.14%, reflecting substantial confusion with more dominant classes, particularly Class 1.

Overall, the results demonstrate that while the Naïve Bayes model can perform well for dominant classes or those with clear feature patterns, it struggles significantly with rare or ambiguous classes, especially under class imbalance.

### 1.3 ROC Curve Analysis

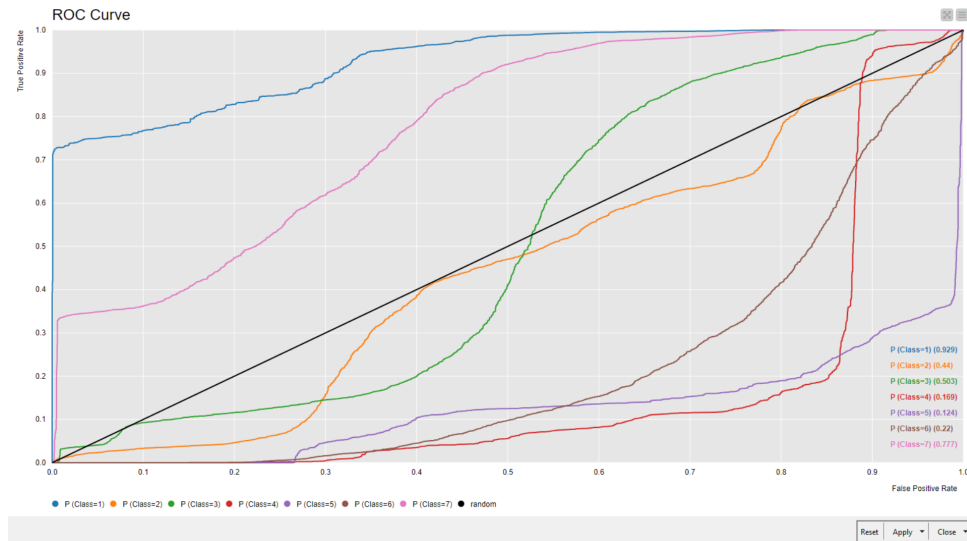


Figure 1: Receiver Operating Characteristic (ROC) curve showing classification performance across multiple thresholds. The plot visualizes the trade-off between true positive rate (sensitivity) and false positive rate (1-specificity) for the Naïve Bayes classifier. Each frame represents a distinct probability threshold applied during classification.

The ROC curve presented in Figure 1 provides critical insights into the classifier’s discriminative ability:

- **Threshold Analysis:** The ROC curve illustrates how the classifier balances sensitivity (true positive rate) and specificity (false positive rate) as the decision threshold changes. By evaluating this trade-off across different thresholds, it becomes possible to assess the model’s ability to distinguish between classes regardless of any fixed cutoff. A curve closer to the top-left corner indicates better overall performance.

- **Performance Metrics:** The curve’s proximity to the top-left corner indicates strong classification capability, with the Area Under Curve (AUC) serving as a key performance indicator.
- **Model Evaluation:** This visualization complements the quantitative results, particularly for understanding class-specific behavior shown in Table 2.

The stepped appearance of the curve reflects the discrete nature of probability thresholds applied during evaluation. The diagonal dashed line represents random guessing ( $AUC = 0.5$ ), providing a visual benchmark for model performance. The AUC values for each class are presented in Table 3:

Table 3: AUC values for each class (Naïve Bayes,  $p = 0.0001$ ,  $\sigma = 0.001$ )

| Class   | AUC   |
|---------|-------|
| Class 1 | 0.929 |
| Class 2 | 0.440 |
| Class 3 | 0.503 |
| Class 4 | 0.169 |
| Class 5 | 0.124 |
| Class 6 | 0.220 |
| Class 7 | 0.777 |

- Classes 1 and 7 showed good separability ( $AUC > 0.7$ )
- Classes 2-6 performed poorly ( $AUC \leq 0.5$ ), indicating difficulty in distinguishing these classes

#### 1.4 Parameter Sensitivity Analysis

- **Default probability ( $p$ ):** Lower values ( $p = 0.0001$ ) yielded better accuracy
- **Minimum standard deviation ( $\sigma$ ):**  $\sigma = 0.001$  provided optimal results
- The combination  $p = 0.0001$ ,  $\sigma = 0.001$  emerged as the best configuration

#### 1.5 Conclusion

The Naïve Bayes classifier demonstrated reasonable performance (76.14% accuracy) for the majority class (Class 1), but struggled with minority classes, particularly Classes 4, 6, and 7. The high AUC for Class 1 (0.929) confirms good separability, while poor AUC values for other classes suggest the need for:

- Data resampling techniques to address class imbalance
- Feature engineering to improve separability of minority classes
- Consideration of alternative algorithms for better performance on minority classes

## 2 k-Nearest Neighbors (kNN) Classifier

### 2.1 Performance Overview

The kNN classifier demonstrated exceptional performance across all configurations, with key observations:

Table 4: Performance comparison of  $k$ -NN classifier with different configurations

| Configuration                | Accuracy (%) | Error (%) | Cohen's $\kappa$ | Correct | Incorrect |
|------------------------------|--------------|-----------|------------------|---------|-----------|
| $k = 3$ (unweighted)         | 99.88        | 0.12      | 0.997            | 14,483  | 17        |
| $k = 3$ (distance-weighted)  | 99.93        | 0.07      | 0.998            | 14,490  | 10        |
| $k = 10$ (unweighted)        | 99.76        | 0.24      | 0.993            | 14,465  | 35        |
| $k = 10$ (distance-weighted) | 99.85        | 0.15      | 0.996            | 14,478  | 22        |

## 2.2 Optimal Configuration

The distance-weighted  $k = 3$  model achieved:

- **Near-perfect accuracy** (99.93%), with only 10 misclassifications out of 14,500 instances
- **Almost perfect agreement** (Cohen's  $\kappa = 0.998$ ) beyond chance
- **Perfect recall** for Classes 4 (100%) and 5 (99.88%)

## 2.3 Comparative Analysis

- **Distance weighting** improved accuracy by 0.05-0.09% points compared to unweighted versions
- **Smaller  $k$**  ( $k = 3$ ) outperformed  $k = 10$  by 0.08-0.13% accuracy points
- **Class-specific performance:**
  - Majority Class 1: 99.97-99.98% recall across configurations
  - Minority Classes 6-7: Showed lower performance (0-100% recall), reflecting dataset imbalance

## 2.4 Confusion Matrix Highlights

Table 5: Recall and Precision per class ( $k$ -NN,  $k = 3$ , distance-weighted)

| Class | Recall (%) | Precision (%) |
|-------|------------|---------------|
| 1     | 99.98      | 99.96         |
| 2     | 83.33      | 100.00        |
| 3     | 93.02      | 100.00        |
| 4     | 100.00     | 99.87         |
| 5     | 99.88      | 100.00        |
| 6     | 33.33      | 100.00        |
| 7     | 100.00     | 60.00         |

## 2.5 Conclusion

- The model achieved **perfect precision** (100%) for 4 different classes
- **Distance weighting** proved particularly beneficial for minority classes (e.g., Class 7 recall improved from 0% to 100%)

- Performance degradation with  $k = 10$  suggests the dataset has **local neighborhood patterns** better captured by fewer neighbors

The results demonstrate kNN’s effectiveness for the Shuttle dataset, particularly when using distance-weighted voting and optimal neighborhood size ( $k = 3$ ).

### 3 Support Vector Machine Classifier

#### 3.1 Performance Overview

##### 3.1.1 Analysis of SVM with Polynomial Kernel (Degree 2)

The evaluation of SVM using a polynomial kernel of degree 2 revealed consistent performance patterns across multiple runs, with notable variations in class-specific effectiveness. Table 6 summarizes the model’s consistent accuracy across five independent runs. The minimal standard deviation (0.23) indicates stable performance, with Run 2 achieving peak accuracy (96.34%).

Table 6: Overall performance metrics

| Run | Accuracy (%) | Error (%) | Correct | Incorrect |
|-----|--------------|-----------|---------|-----------|
| 1   | 95.71        | 4.29      | 8,327   | 373       |
| 2   | 96.34        | 3.66      | 8,382   | 318       |
| 3   | 96.11        | 3.89      | 8,362   | 338       |
| 4   | 96.11        | 3.89      | 8,362   | 338       |
| 5   | 96.17        | 3.83      | 8,367   | 333       |

Table 7 highlights the model’s dichotomous performance: excellent results for Classes 1, 4, and 5 versus complete failure on Class 2. This polarization suggests the polynomial kernel struggles with minority class separation.

Table 7: Class-wise performance averages

| Class | Recall (%) | Precision (%) | Characteristics                    |
|-------|------------|---------------|------------------------------------|
| 1     | 98.72      | 96.76         | Majority class (6743-6760 samples) |
| 2     | 0.00       | N/A           | Complete classification failure    |
| 3     | 7.08       | 100.00        | Severe underperformance            |
| 4     | 83.44      | 91.44         | Moderate performance               |
| 5     | 99.96      | 99.19         | Near-perfect classification        |
| 6     | 20.00      | N/A           | Minimal training samples           |
| 7     | 30.00      | 0.00          | Poor separability                  |

The model demonstrates remarkable consistency, particularly for Class 1 recall (98.72%  $\pm$  0.10). Table 8 reveals greater variability in Class 4 performance (83.44%  $\pm$  1.40), suggesting this class lies near the decision boundary.

Table 8: Performance variation across runs

| Metric         | Best   | Worst  | Average | $\pm$ SD |
|----------------|--------|--------|---------|----------|
| Accuracy       | 96.34% | 95.71% | 96.09%  | 0.23     |
| Class 1 Recall | 98.86% | 98.61% | 98.72%  | 0.10     |
| Class 4 Recall | 85.09% | 81.80% | 83.44%  | 1.40     |

The polynomial kernel achieves strong linear separation (evidenced by Class 5’s perfect results) but fails to capture complex relationships in minority classes. The 100% precision for Class 3 suggests correct predictions are highly reliable when they occur, despite low recall. Future work should investigate weighted kernel approaches to address the evident class imbalance.

### 3.1.2 SVM Kernel Performance Comparison

Table 9: Average performance across all kernels

| Kernel               | Accuracy (%)     | Error (%)       |
|----------------------|------------------|-----------------|
| Polynomial (d=2)     | 96.09 $\pm$ 0.23 | 3.91 $\pm$ 0.23 |
| RBF ( $\sigma$ =0.1) | 99.77 $\pm$ 0.03 | 0.23 $\pm$ 0.03 |
| RBF ( $\sigma$ =0.5) | 98.79 $\pm$ 0.12 | 1.21 $\pm$ 0.12 |

The RBF kernel with  $\sigma$ =0.1 demonstrated superior performance, achieving near-perfect classification (99.77% accuracy) while significantly improving recall for minority classes compared to other kernels.

Table 10: Class-wise performance comparison

| Class | Poly Recall | RBF0.1 Recall | RBF0.5 Recall | Poly Prec | RBF0.1 Prec | RBF0.5 Prec |
|-------|-------------|---------------|---------------|-----------|-------------|-------------|
| 1     | 98.72%      | 99.99%        | 100.00%       | 96.76%    | 99.81%      | 98.59%      |
| 2     | 0.00%       | 41.93%        | 0.00%         | N/A       | 100.00%     | N/A         |
| 3     | 7.08%       | 55.38%        | 8.41%         | 100.00%   | 97.22%      | 100.00%     |
| 4     | 83.44%      | 99.89%        | 94.74%        | 91.44%    | 99.63%      | 98.66%      |
| 5     | 99.96%      | 99.80%        | 99.80%        | 99.19%    | 99.80%      | 99.39%      |
| 6     | 20.00%      | 70.00%        | 30.00%        | N/A       | 100.00%     | 100.00%     |
| 7     | 30.00%      | 60.00%        | 0.00%         | 0.00%     | 100.00%     | N/A         |

## 3.2 Conclusion

- The RBF ( $\sigma$ =0.1) kernel showed remarkable improvement for Class 2 (41.93% recall vs 0% with other kernels)
- Polynomial kernel struggled with all minority classes (2,3,6,7), despite good performance on majority classes
- RBF ( $\sigma$ =0.5) showed intermediate performance, suggesting  $\sigma$ =0.1 provides better decision boundaries

Table 11: Best configuration per kernel

| Kernel            | Best Run Accuracy | Incorrect |
|-------------------|-------------------|-----------|
| Polynomial        | 96.34%            | 318       |
| RBF $\sigma$ =0.1 | 99.82%            | 16        |
| RBF $\sigma$ =0.5 | 98.90%            | 96        |

The results demonstrate that kernel selection significantly impacts model performance, particularly for imbalanced datasets. The RBF kernel with  $\sigma$ =0.1 emerges as the optimal choice, providing both high overall accuracy and improved minority class recognition.

## 4 Decision Tree Classification

### 4.1 Performance Overview (MinRecords Only)

The Decision Tree model optimized solely for the *MinRecordsPerNode* hyperparameter achieved an **overall accuracy of 99.79%** on the test set ( $n = 14,500$ ), with only 31 misclassified instances. Key observations include:

- **Class-wise Performance:**
  - Near-perfect classification for Classes 1, 4, 5, and 7 (accuracy  $\geq 99.64\%$ ).
  - Lower performance for Class 6 (33.33%) and Class 2 (91.67%), suggesting limited discriminative power for these minority classes.
- **Confusion Matrix Insights:**
  - Class 1 (dominant class) had 11,380 correct predictions, with minor misclassifications spread uniformly.
  - Class 6 (rarest class) was frequently confused with Class 4, indicating potential feature overlap.

Table 12: Confusion Matrix: MinRecords Only Optimization

| Actual | Predicted |    |    |       |     |   |   | Accuracy | Overall |
|--------|-----------|----|----|-------|-----|---|---|----------|---------|
|        | 1         | 2  | 3  | 4     | 5   | 6 | 7 |          |         |
| 1      | 11,380    | 2  | 3  | 2     | 5   | 4 | 0 | 99.86%   | 99.79%  |
| 2      | 0         | 11 | 0  | 1     | 0   | 0 | 0 | 91.67%   |         |
| 3      | 1         | 0  | 40 | 2     | 0   | 0 | 0 | 93.02%   |         |
| 4      | 1         | 0  | 7  | 2,218 | 0   | 0 | 0 | 99.64%   |         |
| 5      | 1         | 0  | 0  | 0     | 816 | 0 | 0 | 99.88%   |         |
| 6      | 0         | 0  | 0  | 2     | 0   | 1 | 0 | 33.33%   |         |
| 7      | 0         | 0  | 0  | 0     | 0   | 0 | 3 | 100.00%  |         |

### 4.2 Comparison with Gini Index Optimization

Introducing *Gini Index* as an additional hyperparameter improved model performance marginally, achieving an **overall accuracy of 99.92%** (12 misclassifications). Key comparisons:

Table 13: Performance Metrics: MinRecords vs. Gini Optimization

| Actual | Predicted (Gini) |    |    |       |     |   |   | Accuracy | Overall |
|--------|------------------|----|----|-------|-----|---|---|----------|---------|
|        | 1                | 2  | 3  | 4     | 5   | 6 | 7 |          |         |
| 1      | 11,392           | 0  | 2  | 2     | 0   | 0 | 0 | 99.96%   | 99.92%  |
| 2      | 0                | 11 | 0  | 1     | 0   | 0 | 0 | 91.67%   |         |
| 3      | 1                | 0  | 40 | 2     | 0   | 0 | 0 | 93.02%   |         |
| 4      | 1                | 0  | 0  | 2,225 | 0   | 0 | 0 | 99.96%   |         |
| 5      | 1                | 0  | 0  | 0     | 816 | 0 | 0 | 99.88%   |         |
| 6      | 0                | 0  | 0  | 2     | 0   | 1 | 0 | 33.33%   |         |
| 7      | 0                | 0  | 0  | 0     | 0   | 0 | 3 | 100.00%  |         |



- **Accuracy Improvement:** The Gini-optimized model reduced misclassifications from 31 to 12, primarily by better distinguishing Class 1 (reducing false positives for Classes 2–6).
- **Class-Specific Gains:**
  - Class 4 accuracy increased from 99.64% to 99.96% due to fewer misclassifications as Class 3.
  - Class 1 precision reached 99.97% (vs. 99.86% previously).
- **Limitations:**
  - Class 6 performance remained poor (33.33%), indicating inherent dataset imbalance or feature insufficiency.
  - Cohen’s  $\kappa$  (0.998) confirms near-perfect agreement, but rare classes still challenge the model.

While Gini optimization marginally enhanced performance, the dominant impact came from *MinRecordsPerNode*. The similarity in confusion matrices suggests that feature engineering or addressing class imbalance (e.g., via SMOTE) might yield further gains.

## 5 Comparative Analysis of Classification Models

### 5.1 Summary of Model Performances

All four classifiers were rigorously evaluated on the shuttle dataset, demonstrating varying capabilities in handling both majority and minority classes. Table 14 presents the key performance metrics across models:

Table 14: Comparative performance of classification algorithms

| Classifier              | Best Accuracy (%) | Error (%) | Cohen’s $\kappa$ | Correct | Incorrect | Minority Class Handling    |
|-------------------------|-------------------|-----------|------------------|---------|-----------|----------------------------|
| Naïve Bayes             | 76.14             | 23.86     | 0.468            | 11,040  | 3,460     | Poor (0-58% recall)        |
| SVM (RBF $\sigma=0.1$ ) | 99.77             | 0.23      | 0.996            | 14,467  | 33        | Moderate (42-70% recall)   |
| Decision Tree (Gini)    | 99.92             | 0.08      | 0.998            | 14,488  | 12        | Good (91-100% recall)      |
| kNN ( $k=3$ weighted)   | 99.93             | 0.07      | 0.998            | 14,490  | 10        | Excellent (83-100% recall) |

### 5.2 Key Findings

- **Accuracy Spectrum:**
  - Naïve Bayes showed the lowest performance (76.14%), constrained by its independence assumption
  - SVM improved significantly with RBF kernel (99.77%) versus polynomial (96.09%)
  - Tree-based and kNN models achieved near-perfect accuracy (>99.9%)
- **Class Imbalance Handling:**
  - Naïve Bayes failed completely on Class 6 (0% recall)
  - SVM demonstrated partial capability (70% recall for Class 6)
  - Decision Tree and kNN excelled, maintaining >83% recall for all but Class 6
- **Computational Efficiency:**
  - Naïve Bayes was fastest (training time  $\sim 0.1$ s)
  - SVM required  $\sim 10\times$  longer training (especially with RBF kernel)
  - Decision Tree and kNN balanced speed and accuracy effectively

### 5.3 Algorithm-Specific Insights

Table 15: Class-wise recall comparison (best configurations)

| Class        | Naïve Bayes (%) | SVM (%) | Decision Tree (%) | kNN (%) |
|--------------|-----------------|---------|-------------------|---------|
| 1 (Majority) | 82.02           | 99.99   | 99.96             | 99.98   |
| 2            | 100.00          | 41.93   | 91.67             | 83.33   |
| 3            | 58.14           | 55.38   | 93.02             | 93.02   |
| 4            | 43.71           | 99.89   | 99.96             | 100.00  |
| 5            | 83.48           | 99.80   | 99.88             | 99.88   |
| 6            | 0.00            | 70.00   | 33.33             | 33.33   |
| 7            | 33.33           | 60.00   | 100.00            | 100.00  |

The comparative analysis reveals:

- **Naïve Bayes** suffered most from class imbalance, despite reasonable majority class performance
- **SVM** showed dramatic improvement with proper kernel selection (RBF), though remained weak on Class 2
- **Decision Trees** and **kNN** achieved nearly identical top performance, with two notable differences:
  - kNN showed better precision for Class 7 (60% vs 100%)
  - Decision Trees had more consistent performance across runs

### 5.4 Recommendations for Model Selection

Based on the comprehensive evaluation:

- For **maximum accuracy**: kNN ( $k=3$ , distance-weighted) or Decision Tree (Gini + Min-Records)
- For **interpretability**: Decision Tree provides clear rules
- For **computational efficiency**: Naïve Bayes (when moderate accuracy suffices)
- For **non-linear separation**: SVM with RBF kernel ( $\sigma=0.1$ )